

Operating system

Group-A

1. What is a Process?

A process is a program under execution. The value of the program counter (PC) indicates the address of the next instruction of the process being executed. Each process is represented by a Process Control Block (PCB).

2. What does PCB contain?

A process control block (PCB) contains information about the process, i.e. registers, quantum, priority, etc.

3. What do you mean by inter-process communication?

Inter-process communication (IPC) is a mechanism that allows the exchange of data between processes. By providing a user with a set of programming interfaces, IPC helps a programmer organize the activities among different processes.

4. What is the maximum length of the filename in DOS? 8

5. What else is a command interpreter called? Kernel

6. What is deadlock?

A situation where a set of processes are blocked because each process is holding a resource and waiting for another resource acquired by some other process

7. Define Seek Time and Latency Time.

- A disk is divided into many circular tracks. **Seek Time** is defined as the time required by the read/write head to move from one track to another.
- The disk is divided into many circular tracks, and these tracks are further divided into blocks known as sectors. The time required by the read/write head to rotate to the requested sector from the current position is called **Rotational Latency**.

8. Define waiting time.

- Completion Time (CT): This is the time when the process completes its execution. Arrival Time (AT): This is the time when the process has arrived in the ready state.
- Waiting Time (WT): The time spent by a process waiting in the ready queue for getting the CPU.
- Burst Time (BT): This is the time required by the process for its execution.
- Total Turn Around Time (TAT): The time since the process entered into ready queue for execution till the process completed its execution.

9. When does page fault occur?

A page fault occurs when a program attempts to access data or code that is in its address space, but is not currently located in the system RAM.

10. What is the fence register used for?

The fence register is used for memory protection on the computer. It is a way to access the memory in the computer.

11. Degree of Multiprogramming

The number of processes that can reside in the ready state at maximum decides the degree of multiprogramming, e.g., if the degree of programming = 100, this means 100 processes can reside in the ready state at maximum.

12. What is the use of Paging in an Operating System?

Paging is a memory management technique that is used to retrieve processes from secondary storage to main memory.

13. What is the basic advantage of Paging?

The basic advantage of Paging is that it reduces external fragmentation, but it is not able to reduce internal fragmentation.

14. What is critical-section problem?

The Critical Section Problem:

1. Critical Section – The portion of the code in the program where shared variables are accessed and/or updated.

2. Remainder Section – The remaining portion of the program excluding the Critical Section.

3. Race around Condition – The final output of the code depends on the order in which the variables are accessed. This is termed as the race around condition.

15. What is Demand paging?

specifies that if an area of memory is not currently being used, it is swapped to disk to make room for an application's need.

16. What is Time Sharing?

Time-sharing systems require interaction with the user to instruct the OS to perform various tasks. The OS responds with an output. The instructions are usually given through an input device like the keyboard.

17. What are operating system service?

- Services of Operating System
- Program execution
- Input Output Operations
- Communication between Process
- File Management
- Memory Management
- Process Management
- Security and Privacy
- Resource Management
- User Interface
- Networking
- Error handling
- Time Management

18. What is Kernel?

Kernel is central component of an operating system that manages operations of computer and hardware. It basically manages operations of memory and CPU time. It is core component of an operating system. Kernel acts as a bridge between applications and data processing performed at hardware level using inter-process communication and system calls.

19. What are privileged and non- privileged instructions?

- **privileged instructions** are those that can only be executed by the operating system kernel or a privileged process, such as a device driver. These instructions typically perform operations that require direct access to hardware or other privileged resources, such as setting up memory mappings or accessing I/O devices. Privileged instructions are executed in kernel mode, which provides unrestricted access to the system resources.
- **Non-privileged instructions** are those that can be executed by any process, including user-level processes. These instructions are typically used for performing computations, accessing user-level resources such as files and memory, and managing process control. Non-privileged instructions are executed in user mode, which provides limited access to system resources and ensures that processes cannot interfere with one another.

20. What is meant by Context Switch?

Context switching in an operating system involves saving the context or state of a running process so that it can be restored later, and then loading the context or state of another process and run it.

21. Need of context switch?

A context switch can also occur as the result of an interrupt, when a task needs to access disk storage, freeing up CPU time for other tasks. The process of context switching can have a negative impact on system performance.

22. What is the use of fork system call?

The fork system call in an operating system is used to create a new process by duplicating the existing process.

23. What is starvation?

Starvation is Resource management problem. In this problem, a waiting process does not get the resources it needs for a long time because the resources are being allocated to other processes.

24. Define logical address.

A logical address in operating systems is a virtual address generated by a process during its execution.

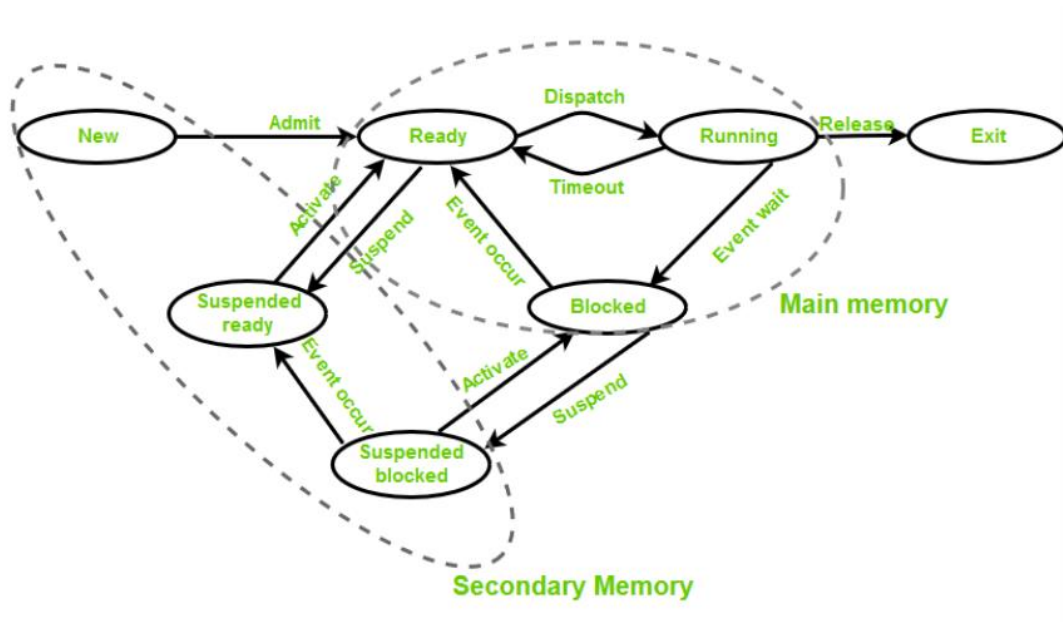
Group-B

1A) Explain lifecycle of a process with a suitable diagram

The states of a process are as follows:

1. **New (Create):** In this step, the process is about to be created but not yet created. It is the program that is present in secondary memory that will be picked up by OS to create the process.
2. **Ready:** New -> Ready to run. After the creation of a process, the process enters the ready state i.e. the process is loaded into the main memory. The process here is ready to run and is waiting to get the CPU time for its execution. Processes that are ready for execution by the CPU are maintained in a queue called ready queue for ready processes.
3. **Run:** The process is chosen from the ready queue by the CPU for execution and the instructions within the process are executed by any one of the available CPU cores.
4. **Blocked or Wait:** Whenever the process requests access to I/O or needs input from the user or needs access to a critical region (the lock for which is already acquired) it enters the blocked or waits for the state. The process continues to wait in the main memory and does not require CPU. Once the I/O operation is completed the process goes to the ready state.
5. **Terminated or Completed:** Process is killed as well as PCB is deleted. The resources allocated to the process will be released or deallocated.

6. **Suspend Ready:** Process that was initially in the ready state but was swapped out of main memory(refer to Virtual Memory topic) and placed onto external storage by the scheduler is said to be in suspend ready state. The process will transition back to a ready state whenever the process is again brought onto the main memory.
7. **Suspend wait or suspend blocked:** Similar to suspend ready but uses the process which was performing I/O operation and lack of main memory caused them to move to secondary memory. When work is finished it may go to suspend ready.



1B) What is PCB and explain it.

A process control block (PCB) is a data structure used by operating systems to store important information about running processes. It contains information such as the unique identifier of the process (Process ID or PID), current status, program counter, CPU registers, memory allocation, open file descriptions and accounting information. The circuit is critical to context switching because it allows the operating system to efficiently manage and control multiple processes.

3. A) Differentiate between a Process and a Program.

Program	Process
Program contains a set of instructions designed to complete a specific task.	Process is an instance of an executing program.
Program is a passive entity as it resides in the secondary memory.	Process is a active entity as it is created during execution and loaded into the main memory.
Program exists at a single place and continues to exist until it is deleted.	Process exists for a limited span of time as it gets terminated after the completion of task.
Program is a static entity.	Process is a dynamic entity.
Program does not have any resource requirement, it only requires memory space for storing the instructions.	Process has a high resource requirement, it needs resources like CPU, memory address, I/O during its lifetime.

Program does not have any control block.	Process has its own control block called Process Control Block.
Program has two logical components: code and data.	In addition to program data, a process also requires additional information required for the management and execution.
Program does not change itself.	Many processes may execute a single program. There program code may be the same but program data may be different. these are never same.
Program contains instructions	Process is a sequence of instruction execution.

3.B) Differentiate distributed systems from multiprocessor system?

Multiprocessor VS Distributed

Based on	Multiprocessor	Distributed
Scalability	Harder	Easier
Programmability	Easier	Harder
Degree of transparency	High	Very high
Architecture	Homogenous	Heterogenous
Basic of communication	Shared memory	Message passing

4. A) What is scheduler?

Process scheduling is the activity of the process manager that handles the removal of the running process from the CPU and the selection of another process based on a particular strategy.

4. B) What are different types of schedulers? Explain it.

Long-term – performance: Decides how many processes should be made to stay in the ready state. This decides the degree of multiprogramming. Once a decision is taken it lasts for a long time which also indicates that it runs infrequently. Hence it is called a long-term scheduler.

Short-term – Context switching time: Short-term scheduler will decide which process is to be executed next and then it will call the dispatcher. A dispatcher is a software that moves the process from ready to run and vice versa. In other words, it is context switching. It runs frequently. Short-term scheduler is also called CPU scheduler.

Medium-term – Swapping time: Suspension decision is taken by the medium-term scheduler. The medium-term scheduler is used for swapping which is moving the process from main memory to secondary and vice versa. The swapping is done to reduce degree of multiprogramming.

5. A) Define independent process and dependent process.

- **Independent Process:** Independent Processes are those processes whose task is not dependent on any other processes.
- **Cooperating Process:** Cooperating Processes are those processes that depend on other processes or processes. They work together to achieve a common task in an operating system. These processes interact with each other by sharing the resources such as CPU, memory, and I/O devices to complete the task.

B) What is IPC? Name different methods of it.

Inter-process communication (IPC) is a mechanism that allows the exchange of data between processes. By providing a user with a set of programming interfaces, IPC helps a programmer organize the activities among different processes.

- a. **Pipes:** One-way communication channels between related or unrelated processes.
- b. **Message Passing:** Processes communicate through messages, either directly or through mailboxes.
- c. **Shared Memory:** Processes share a portion of their memory space for fast communication.

6. Explain about resource allocation graph (RAG)?

A Resource Allocation Graph (RAG) is a graphical representation used in deadlock detection algorithms to model the resource allocation relationships among processes in a system. It is a directed graph that depicts how resources are allocated and requested by processes. The purpose of the resource allocation graph is to identify potential deadlocks in a system.

Here are the key components and explanations related to a Resource Allocation Graph:

Nodes: Nodes in a Resource Allocation Graph represent two types of entities: processes and resources.

Circle (Oval): Represents a process.

Square (Box): Represents a resource.

Edges: Edges represent relationships between processes and resources in terms of resource allocation and resource request.

Arrow from a process to a resource: Indicates that the process is currently allocated the resource.

Arrow from a resource to a process: Indicates that the process is currently requesting the resource.

Resource Types: Different types of resources are represented by distinct nodes in the graph. Each resource type can have multiple instances, and these instances are represented by individual nodes of that type.

Resource Instances: Nodes representing resource instances contain information about the units available and the units currently held by a process.

Wait-for-Graph Aspect: In addition to representing resource allocation, a Resource Allocation Graph can be extended to include the wait-for-graph aspect. If a process is waiting for a resource held by another process, this relationship is represented by a dashed line between the two processes.

Deadlock Detection: Deadlocks in a system can be detected by analyzing cycles in the Resource Allocation Graph. If a cycle contains only resource-request edges and no resource-allocation edges, it indicates a potential deadlock situation. The processes involved in the cycle form a set of processes in a deadlock.

Deadlock Resolution: Once a deadlock is detected, the system can take corrective actions to resolve it. This may involve preempting resources from some processes, rolling back the execution of certain processes, or releasing resources to break the deadlock cycle.

Resource Allocation Graphs are particularly useful in scenarios where multiple processes contend for a finite set of resources. By visualizing the allocation and request relationships, system administrators can identify potential deadlocks and implement strategies to prevent or resolve them.

7. A) What are the advantages of Contiguous Allocation?

Both the Sequential and Direct Accesses are supported by this. For direct access, the address of the k th block of the file which starts at block b can easily be obtained as $(b+k)$.

This is extremely fast since the number of seeks are minimal because of contiguous allocation of file blocks.

7. B) How protection is done during paging?

Page Tables and Page Table Entries : The operating system uses page tables to map virtual pages to physical frames. Each Page Table Entry (PTE) within the page table contains flags or bits that control access rights and permissions, such as Valid/Invalid, Read/Write, User/Supervisor, and Execute/No-Execute bits.

Access Control at the Page Level: Protection is achieved by setting permissions at the page level. Memory pages can be marked as read-only, read-write, execute-only, etc. This provides fine-grained control over access rights for individual pages.

Handling Access Violations : If a process attempts to access a page without the necessary permissions, a page fault occurs. The operating system catches this exception, checks the access violation, and takes appropriate action, which may include terminating the process or handling the error according to the specified program behavior.

8. Define OS. List the objective of OS. Explain how multiprogramming increase CPU utilization?

An Operating System can be defined as an interface between user and hardware. It is responsible for the execution of all the processes, Resource Allocation, CPU

management, File Management and many other tasks. The purpose of an operating system is to provide an environment in which a user can execute programs in a

convenient and efficient manner.

OBJECTIVES OF OS

- **Convenience:** The operating system aims to provide a user-friendly and convenient interface for interacting with the computer making it easier for users to navigate and utilize the system.
- **Efficiency:** Operating systems strive to optimize resource utilization and system performance.
- **Scalability:** Scalability is a crucial objective, allowing the operating system to adapt to changes in system size and workload, OS can manage growth seamlessly without significant degradation in performance.
- **Robustness:** implies the ability of the operating system to handle errors, faults, and unexpected situations gracefully.
- **Reliability:** Users rely on the OS to execute tasks correctly, manage data reliably, and maintain system integrity.
- **Portability:** involves the ability of the operating system to run on different hardware architectures or platforms.

Multiprogramming is a technique used in operating systems to increase CPU utilization and enhance overall system efficiency.

Overlap of CPU and I/O Operations: In a multiprogramming environment, multiple programs or processes are loaded into the system's memory simultaneously hence increasing throughput.

Context Switching: Multiprogramming involves quick context switching between different processes. When one process is waiting for I/O, the operating system can quickly switch to another process that is ready to execute.

9 A) What is synchronization?

Process Synchronization is the coordination of execution of multiple processes in a multi-process system to ensure that they access shared resources in a controlled and predictable manner. It aims to resolve the problem of race conditions and other synchronization issues in a concurrent system.

B) Give Peterson's solution to the Critical section problem.

Peterson's solution is a classic algorithm used to solve the critical section problem, which involves providing mutual exclusion to processes that share a common resource. Here's a brief explanation for a 3-mark answer:

Two-Process Mutual Exclusion: Peterson's solution is designed for the scenario where only two processes need to access a critical section of code concurrently. The solution guarantees mutual exclusion, ensuring that only one process can execute the critical section at a time.

Key Data Structures: The algorithm relies on two shared variables: flag and turn. Each process has its own flag variable, and the turn variable is used to determine whose turn it is to enter the critical section. These variables are typically implemented as Boolean flags.

Algorithm Steps: The key steps in Peterson's solution involve setting flags and checking turns. Here's a high-level overview:

- Each process sets its flag to indicate its intention to enter the critical section.
- The process also sets the turn variable to the other process, indicating that it is the other process's turn.
- Before entering the critical section, a process checks if the other process has set its flag and if it is the other process's turn. If both conditions are true, the process enters the critical section; otherwise, it waits.
- After exiting the critical section, the process resets its flag, allowing the other process to enter.

Peterson's solution is a simple yet effective way to provide mutual exclusion for two processes. However, it may not scale well to more than two processes, and more complex solutions, such as the use of semaphores or mutexes, are often preferred for larger scenarios

10 A) Distinguish between Logical and Physical address space.

Logical Address	Physical Address
It is an address generated by the CPU during program execution.	It refers to a physical location in the memory unit.
User programs deal with the logical address directly.	The user program never sees the physical address.
The logical address is referred to as the physical address.	Users cannot directly access the physical address.
The set of all logical addresses generated by a program are known as the 'logical address space'.	The set of all physical addresses is known as the 'physical address space'.
A logical address does not exist physically and is referred to as a 'virtual address'.	A physical address is a real location that exists in the memory unit.

B) What do you mean by First Fit and best fit memory allocation method?

First Fit: First Fit is a memory allocation algorithm that assigns the first available block of memory that is large enough to accommodate a process. It searches the memory from the beginning and allocates the first block that satisfies the size requirement of the process.

Best Fit: Best Fit is a memory allocation algorithm that selects the smallest available block of memory that is large enough to fit a process. It aims to minimize the wastage of memory by choosing the most suitable block from the available options.

Worst Fit: Worst Fit is a memory allocation algorithm that allocates the largest available block of memory to a process. The idea is to maximize the utilization of memory and potentially reduce fragmentation, but it may lead to inefficient use of space over time.

Next Fit: Next Fit is a variation of the First Fit algorithm where the search for an available block starts from the last allocated position. It continues searching for a suitable block until it finds one or completes a full cycle through the memory. This algorithm can help reduce fragmentation compared to pure First Fit.

11 A) Consider a logical address space of 64 pages of 1024 words each, mapped onto a physical memory of 32 frames.

i) How many bits are there in the logical address?

ii) How many bits are there in the physical address?

To calculate the number of bits in the logical and physical addresses, we need to determine the size of the logical address space and the physical memory.

Given:

- Logical address space: 64 pages
- Page size: 1024 words
- Physical memory: 32 frames

Let's calculate the bits for each:

i) Logical Address:

- The logical address space is given by the product of the number of pages and the page size.

Logical Address Space = Number of Pages $\times$ Page Size

Logical Address Space = 64 pages $\times$ 1024 words/page

Logical Address Space = 65536 words

Now, to find the number of bits needed to represent this space, we use the formula:

Number of bits = $\log_2(\text{Word Count})$

Number of bits = $\log_2(65536)$

Number of bits = 16 bits

Therefore, there are 16 bits in the logical address.

ii) Physical Address:

- The physical memory size is given by the product of the number of frames and the frame size.

Physical Memory Size = Number of Frames $\times$ Frame Size

Physical Memory Size = 32 frames $\times$ 1024 words/frame

Physical Memory Size = 32768 words

Similarly, to find the number of bits needed to represent this space:

Number of bits = $\log_2(\text{Word Count})$

Number of bits = $\log_2(32768)$

Number of bits = 15 bits

Therefore, there are 15 bits in the physical address.

In summary:

- The logical address has 16 bits.
- The physical address has 15 bits.

Group-C

1.A) Differentiate SCAN and C-SCAN scheduling algorithm.

S. No.	SCAN Scheduling Algorithm	C-SCAN Scheduling Algorithm
1.	It is also known as Elevator Algorithm.	It is also known as Circular Elevator Algorithm.
2.	It services all the requests in both the direction.	It services the requests in one direction only.
3.	In above example of SCAN algorithm, head starts moving from 53 towards left servicing all the requests in that direction and further, reaching at left end, it changes the head direction to right and further servicing all the requests from 0 to 199.	In above example of C-SCAN algorithm, head starts from 53 toward right servicing all the requests in that direction and after reaching at right end it does not reverses its direction rather, head jumps to the opposite end of the disk servicing the requests on its right.
4.	SCAN Algorithm provides a longer waiting time to request the locations.	C-SCAN Algorithm provides uniform waiting time in requesting the locations over Elevator Algorithm.
5.	It has higher throughput and provides low variance response time.	This algorithm provides better response time.

B) What is Belady's Anomaly?

A page fault occurs when a page is not found in the memory and needs to be loaded from the disk. If a page fault occurs and all memory frames have been already allocated, then the replacement of a page in memory is required on the request of a new page. This is referred to as demand-paging. The choice of which page to replace is specified by page replacement algorithms. The commonly used page replacement algorithms are FIFO, LRU, optimal page replacement algorithms, etc.

Generally, on increasing the number of frames to a process' virtual memory, its execution becomes faster as fewer page faults occur. Sometimes the reverse happens, i.e. more page faults occur when more frames are allocated to a process. This most unexpected result is termed Belady's Anomaly.

Bélády's anomaly is the name given to the phenomenon where increasing the number of page frames results in an increase in the number of page faults for a given memory access pattern.

C) Define and explain necessary conditions of deadlock.

Deadlock can arise if following four conditions hold simultaneously (Necessary Conditions):

1. Mutual Exclusion – One or more than one resource is non-sharable (Only one process can use at a time).
2. Hold and Wait – A process is holding at least one resource and waiting for

resources.

3. No Preemption – A resource cannot be taken from a process unless the process releases the resource.

4. Circular Wait – A set of processes are waiting for each other in circular form.

D) Explain about various fields of Process Control Block. What are the advantages of inter-process communication? How communication takes place in a shared-memory environment? Explain.

Process State: The state of the process is stored in the PCB which helps to manage the processes and schedule them. There are different states for a process which are “running,” “waiting,” “ready,” or “terminated.”

Process ID: The OS assigns a unique identifier to every process as soon as it is created which is known as Process ID, this helps to distinguish between processes.

Program Counter: While running processes when the context switch occurs the last instruction to be executed is stored in the program counter which helps in resuming the execution of the process from where it left off.

CPU Registers: The CPU registers of the process helps to restore the state of the process so the PCB stores a copy of them.

Memory Information: The information like the base address or total memory allocated to a process is stored in PCB which helps in efficient memory allocation to the processes.

Process Scheduling Information: The priority of the processes or the algorithm of scheduling is stored in the PCB to help in making scheduling decisions of the OS.

Accounting Information: The information such as CPU time, memory usage, etc helps the OS to monitor the performance of the process.

Advantages of Using Process Control Block

- As, PCB stores all the information about the process so it lets the operating system execute different tasks like process scheduling, context switching, etc.
- Using PCB helps in scheduling the processes and it ensures that the CPU resources are allocated efficiently.
- When the different resource utilization information about a process are used from the PCB they help in efficient resource utilization and resource sharing.
- The CPU registers and stack pointers information helps the OS to save the process state which helps in Context switching.

Advantages of IPC:

- Enables processes to communicate with each other and share resources, leading to increased efficiency and flexibility.
- Facilitates coordination between multiple processes, leading to better overall system performance.
- Allows for the creation of distributed systems that can span multiple computers or networks.
- Can be used to implement various synchronization and communication protocols, such as semaphores, pipes, and sockets.

communication takes place in a shared-memory environment:

Shared Memory Region: In a shared-memory environment, there exists a region of memory that is designated for sharing data among multiple processes. This region is accessible to all participating processes, allowing them to read from and write to the same memory space.

Read and Write: Processes communicate by performing read and write operations on the shared memory. For example, one process can write data into a specific location in the shared memory, and another process can read that data from the same location. This shared memory acts as a medium for inter-process communication.

Synchronization Mechanisms: To avoid conflicts and ensure consistency in data sharing, synchronization mechanisms such as locks, semaphores, or mutexes are often employed. These synchronization tools help control access to shared resources within the shared-memory environment, preventing race conditions and ensuring that processes interact with the shared memory in a coordinated manner.

E) What is Dining Philosophers problem? Discuss the solution to the Dining philosopher's problem using monitors.

States that there are 5 Philosophers who are engaged in two activities Thinking and Eating. Meals are taken communally in a table with five plates and five forks in a cyclic manner as shown in the figure.

Constraints and Condition for the problem:

- Every Philosopher needs two forks in order to eat.
- Every Philosopher may pick up the forks on the left or right but only one fork at once.
- Philosophers only eat when they had two forks. We have to design such a protocol i.e. pre and post protocol which ensures that a philosopher only eats if he or she had two forks.
- Each fork is either clean or dirty.



Solution :

Correctness properties it needs to satisfy are :

- Mutual Exclusion Principle –
- No two Philosophers can have the two forks simultaneously.
- Free from Deadlock –
- Each philosopher can get the chance to eat in a certain finite time.
- Free from Starvation –When few Philosophers are waiting then one gets a chance to eat in a while.

- No strict Alternation.
- Proper utilization of time.

F) Define CPU Scheduling. Distinguish between preemptive and non-preemptive Scheduling. What are the various scheduling criteria for CPU scheduling?

CPU Scheduling is a process that allows one process to use the CPU while another process is delayed (in standby) due to unavailability of any resources such as I / O etc, thus making full use of the CPU. The purpose of CPU Scheduling is to make the system more efficient, faster, and fairer.

Preemptive	Non-preemptive
1. Resources allocated to process for limited time.	1. Resources allotted till process is complete/switches to waiting.
2. Processes can be interrupted.	2. Process cannot be interrupted.
3. Low priority process may starve.	3. Short timed process may starve.
4. Has overheads.	4. No overheads.
5. Flexible.	5. Rigid.
6. Cost associated.	6. Not cost associative.
7. Waiting time, Response time is less	7. Waiting time, Response time is more.

CPU scheduling is a crucial aspect of operating systems, and different scheduling criteria are used to determine the order in which processes are executed on the CPU. Here are various scheduling criteria used for CPU scheduling: CPU Utilization: Maximizing CPU utilization is a common scheduling criterion. The goal is to keep the CPU busy and prevent it from remaining idle to ensure efficient use of resources.

Throughput: Throughput refers to the number of processes completed in a unit of time. Scheduling algorithms aim to maximize throughput by executing and completing a significant number of processes within a given timeframe.

Turnaround Time: Turnaround time is the total time taken to execute a process, including both waiting time and execution time. Scheduling algorithms may aim to minimize turnaround time to improve the overall responsiveness of the system.

Waiting Time: Waiting time is the time a process spends waiting in the ready queue before getting the CPU. Minimizing waiting time is important to enhance the efficiency and responsiveness of the system.

Response Time: Response time is the time it takes for a system to respond to a user's request. Scheduling algorithms often aim to minimize response time to provide a more interactive and user-friendly environment.

Fairness: Fairness involves treating all processes equally and ensuring that each process gets a fair share of the CPU time. Scheduling algorithms aim for fairness to prevent starvation and ensure equitable resource allocation.

Predictability: Predictability is the ability to anticipate and control the behavior of the scheduling algorithm. Some systems prioritize predictability to create a stable and consistent environment, which is essential for real-time systems.

Priority: Priority scheduling assigns priority levels to processes, and the CPU is allocated to the process with the highest priority. Prioritizing certain processes allows the system to meet specific performance or business requirements.

Preemption: Preemption is the ability to stop the execution of a currently running process and allocate the CPU to another process. Preemptive scheduling algorithms allow for efficient handling of higher-priority tasks.

Response Ratio: The response ratio is a criterion that considers both waiting time and the execution time of a process. It is used in some scheduling algorithms to make decisions based on the responsiveness of processes.

Different operating systems and applications may prioritize these criteria differently based on their specific requirements and goals. The choice of a scheduling algorithm depends on the characteristics of the system and the intended use of the computing resources.